The Disappearance of Writing Systems

Dedicated to the memory of Jeremy Black (1951–2004)

The Disappearance of Writing Systems

Perspectives on Literacy and Communication

Edited by

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1

Now You See It; Now You Don't! The Disappearance of the Linear A Script on Crete[1]

John Bennet

Introduction: Background and Issues

This chapter explores the disappearance of one of the three scripts in use in the Aegean world (primarily the island of Crete, the southern mainland of Greece, and some of the Aegean islands) in the second millennium BC.

Arthur Evans, the excavator of the site of Knossos (see map Fig. 1.1), was keen to claim the Minoan 'civilization' of Crete, whose remains he had uncovered, as the first European civilization (e.g. Farnell 1927). One pressing motivation was the fact that Crete had only 'left' the shrinking Ottoman Empire in 1898, two years before Evans began his excavations at Knossos (see Brown 2000; 2001). For this first 'European' civilization to justify its status, it had to have a writing system, like the older 'Oriental' civilizations of Mesopotamia and Egypt. Indeed it was the existence of a so-called 'pictographic' writing system on stone seals acquired by Evans as keeper of Oxford's Ashmolean Museum in the 1880s that first led him to Crete, and they were a major piece of evidence for what Evans described in one of his early publications as a 'prae-Phoenician' writing system on the island (Evans 1895; see Bennet 2000). Another potent influence on Evans was the developing theory of evolution that had been adopted by British prehistorians, notably his own father, John Evans (see e.g. MacGillivray 2000: 30–3; Trigger 2006: 146).

These notions structured Evans' interpretation of cultural development, including that of script (first presented in Evans 1909), influencing the chronological framework he developed for his Minoan civilization. His chronology,

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Figure 1.1 The Aegean, showing sites mentioned in the text.

subsequently modified extensively in detail, was a tripartite scheme, based on a perceived sequence of rise, florescence, and decline, and calibrated against a ceramic sequence loosely related to stratigraphy (Table 1.1). An absolute chronology was established by reference to the Egyptian 'historical' chronology through finds of Egyptian imports in Aegean contexts and of Aegean objects in Egypt (see Warren and Hankey 1989 for a fully documented exposition; also Kemp and Merrillees 1980). In this tripartite sequence, Evans divided the Bronze Age of Crete into three phases—Early, Middle, and Late—and gave it the overall term 'Minoan' (those on the Greek mainland and islands follow similar tripartite divisions, but are termed 'Helladic' and 'Cycladic' respectively). Each broad phase was divided into three—I, II, and III—and these in turn, where appropriate, subdivided into A, B, and C. As it is currently understood, therefore, the Bronze

Age began on Crete with the Early Minoan (EM) phase, continued through the Middle Minoan (MM), and ended with the close of the Late Minoan (LM).

Although Evans' system, with subsequent refinement, allows a fairly precise and narrow phasing of ceramics, major cultural shifts did not follow these neat, arbitrary divisions, necessitating the development of a broader, more descriptive terminology around the appearance of the complex architectural forms called 'palaces' (Table 1.1). Thus, the period covering the Early Minoan and the very beginning of Middle Minoan (phase MMIA), before the construction of the palaces, is called 'Pre-Palatial', while a major shift *within* the palatial phase is marked by the distinction between Proto- and Neo-Palatial. Many palatial and other sites suffered destruction at the end of the Neo-Palatial phase, but Knossos continued as a palatial centre for some generations, to which we now assign the term Final-Palatial (the era of Knossos' Linear B administration), followed by the Post-Palatial (the period after Knossos' destruction). Of greatest relevance to this chapter are the latest Pre-Palatial, when writing first appeared, and the end of the Neo-Palatial / beginning of Final-Palatial (Table 1.1), when Linear A disappears. While the relative chronology and thus the order in which developments occurred are not in dispute, Aegean archaeologists cannot yet agree on the absolute chronology of the first phase of the Late Bronze Age (Late Minoan IA–B). Conventional chronology would place this c. 1600–1450 BC (Warren and Hankey 1989: 137–46, 169, table 3.1), while recent work suggests that the beginning of the phase should be raised to c. 1710 BC, a difference of about a century (Manning *et al.* 2006). The effect of the 'high' chronology is to lengthen the Neo-Palatial phase, suggesting that developments took place over a longer timescale than the 'low' chronology allows (see Table 1.1).

A final element of the intellectual context created by Evans' legacy is the principle of Minoan primacy and supremacy in the Bronze Age Aegean. For Evans, civilization in Europe began on Crete and was brought to the rest of the Aegean by Cretans as a result of their naval supremacy. This analysis dominated the study of Aegean prehistory for the first half of the twentieth century and led to a profound difference of opinion between Evans and Alan Wace who, with Carl Blegen, established the second-millennium-BC cultural sequence on the Greek mainland (e.g. Blegen and Wace 1916–1918; also Wace 1956; Fitton 1995: 150–65). In terms of language, the logical conclusion of Evans' view was that the Minoan civilization, being of greater antiquity than that on the Greek mainland, must have used a language that preceded Greek. For him, this language was in use throughout much of the southern Greek mainland in the later second millennium BC (Evans 1935: 752–5; implicit also in Ventris 1940: esp. 494, 502). (On the contrary, Wace [e.g. 1956: xxiv] claimed that Greek-speakers arrived in mainland Greece at the beginning of the Middle Bronze Age). Logically for Evans, therefore, none of the three scripts he had identified could have notated the Greek language; for him, they more likely notated an 'older' language.

Dates (BC)	Pottery phase	Cultural phase	Script in use
2100–1900	Middle Minoan (MM) IA	Pre-Palatial	Archanes script
1900–1700	MM IB–III	Proto-Palatial	Cretan Hieroglyphic; Linear A
1700–1450	Late Minoan (LM) IA–B	Neo-Palatial	Linear A
1450–1350	LM II–IIIA1	Final-Palatial	Linear B
1350–1200	LMIIIA2–IIIB	Post-Palatial	Linear B

Table 1.1 Outline chronology of Minoan Crete, showing approximate absolute dates, pottery phases, cultural phases, and scripts in use (cf. Warren and Hankey 1989; Manning *et al.* 2006).

It was this understanding that led Michael Ventris to draw on Etruscan, a possible relict 'pre-Greek' language, as a basis for understanding the language behind Linear B (Ventris 1940; cf. Bennett 1989). Equally, it is against this background that we can better understand the comments which he describes as a 'frivolous digression' (Ventris 1988: 327) in his famous 'Work Note 20', immediately prior to his decipherment of Linear B in 1952, which he concludes with the following words (Ventris 1988: 331):

> If pursued, I suspect that this line of decipherment would sooner or later come to an impasse, or dissipate itself in absurdities; and that it would be necessary to revert to the hypothesis of an indigenous, non-Indo-European language. But this fantasy may be the excuse for us once more to ask ourselves the question: Which is historically more incongruous, a Knossos which writes Greek, or a Mycenae which writes 'Cretan'?

The importance of the demonstration that the language behind the Linear B script is Greek is akin to that of Evans' initial claim that Minoan civilization is European: it appears to show that Greek speakers (strictly only 'writers') existed not only on the Greek mainland, but also on the island of Crete, in the later phases of the Aegean Bronze Age. Furthermore, scholars like Wace and Blegen, vindicated in their view of the significance of mainland Greece in this period, interpreted the presence of the Greek language on Crete as evidence of mainland supremacy over Minoan. Some of these issues are made clear in Wace's foreword to the first major edition of a collection of Linear B texts published in 1956 (Wace 1956: xxxi):

> The importance of Mr Ventris' decipherment can hardly be over-estimated, for it inaugurates a new phase in our study of the beginnings of classical Hellas. We must recognize the Mycenaean culture as Greek, and as one of the first stages in the advance of the Hellenes towards the brilliance of their later amazing achievements ... . In culture, in history and in language we must regard prehistoric and historic Greece as one indivisible whole.

Wace's words reveal an essentialist view of identity, in which language is of paramount importance. The 'proof' the decipherment offered that Greek speak-

ers had taken over the Minoan civilization in the later second millennium BC made Evans' slightly dubious claim of Europeanness irrefutable, because prehistoric Greece was now directly linked to classical Greece by the thread of a single language. I would suggest not only that the polarization of the field in the first part of the twentieth century delayed the decipherment of Linear B, but also that its legacy has been a simplistic reconstruction of the changes that took place on Crete, especially at Knossos, in the fifteenth century BC. These have been interpreted as the consequence of a mainland invasion by Mycenaean Greeks, either learning to write upon their arrival, or, in the more extremely 'essentialist' view, bringing their own writing system with them. Among the changes that the fifteenth century BC saw on Crete is the disappearance of Linear A, which is the focus of this chapter.

A Brief History of Writing and Administration in the Aegean

Writing came relatively late to the Aegean world and is attested on Crete some time before its first appearance on the Greek mainland. It seems unlikely that Aegean writing was an independent invention, given the well-documented connections between the region and the eastern Mediterranean world that seem to have become more regular and intense in the later third millennium BC, facilitated by the introduction of the sailing ship (e.g. Broodbank 2000: 341–9; Sherratt 2000: 18–20). However, all three Aegean scripts are structurally distinct from those in neighbouring regions, being simple syllabaries, based on relatively small repertoires of phonetic signs, together with quite large numbers of logographic signs representing commodities. Evans, again in keeping with the intellectual climate of his time, saw the origins of script in Egypt (e.g. Evans 1909: 118–34), and this partly explains his label for its earliest manifestation, namely 'hieroglyphic', also known as 'pictographic'. Furthermore, he saw the development of the Cretan scripts in evolutionary terms, moving from 'pictographic' forms, often with recognizable prototypes in the material world, towards more conventionalized, 'linear' forms (Evans 1909: 8–18; 1952: 1–2). In particular, he identified an early script, now conventionally known as Cretan Hieroglyphic (e.g. Olivier and Godart 1996; Younger 1999; Karnava 2000) and itself divided into progressive variants—A and B (Evans 1909: 134–48)—that was superseded by the first 'linear' form: Linear A. This was replaced by the second—Linear B—but, according to Evans, only at Knossos, where it represented the most sophisticated, palatial variant, in contemporaneous use with the latest Linear A at Neo-Palatial sites of the same period (Evans 1909: 38–9).

Discoveries as well as systematic study and re-analysis since Evans have modified this evolutionary scheme. In the first instance, writing has been recognized on a small set of seals, bearing signs that have collectively become known as the 'Archanes script' (Olivier and Godart 1996: 31; Godart 1999), after the find-spot of

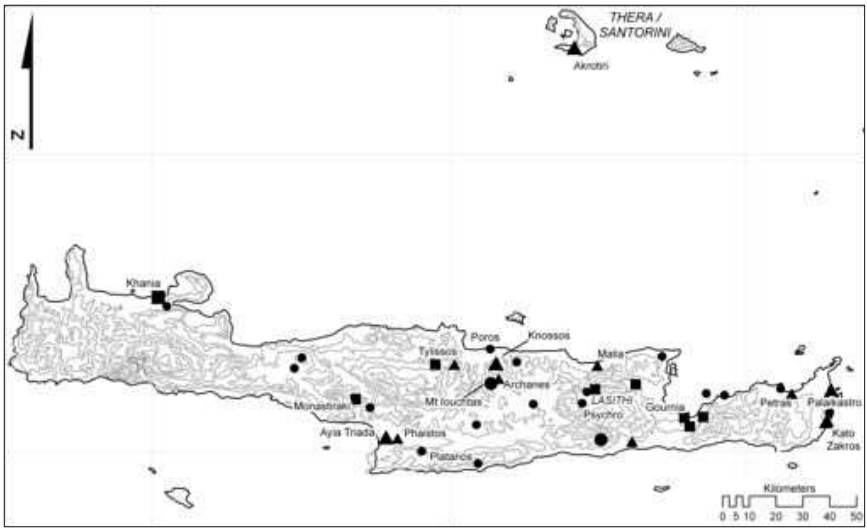

Figure 1.2 Crete and Thera/Santorini, showing sites producing Linear A documents. Sites mentioned in the text are named. Solid squares indicate sites with only administrative documents; solid circles, those with only non-administrative documents; and solid triangles, those with examples of both. Larger symbols indicate greater numbers of attested documents.

examples in the long-lived Fournoi cemetery near Archanes (Fig. 1.2), about 10 km south of Knossos in north-central Crete (Sakellarakis and Sapouna-Sakellaraki 1997: 326–30; Krzyszkowska 2005: 70–1). These belong in the final Pre-Palatial phase (Table 1.1) and therefore push back the history of writing in the region. It has become clear that, rather than constituting a distinct script whose use did not continue, the relatively small number of signs known in the Archanes script are best regarded as the earliest attestations of Cretan Hieroglyphic (Younger 1999: 380–1), itself regularly incised on stone seals, or even of Linear A (Godart 1999). Thus, the first true writing on Crete appears in an elite milieu, apparently unconnected with administration (see Schoep 2006: 44–8). However, we need to be aware of the taphonomy of writing in Crete: we are dependent on writing being applied to materials that are preserved archaeologically, such as stone or bone, *and* on conditions being favourable to the preservation of those materials. Since clay was in wide and regular use for the production of many types of administrative document in the Aegean and these were not deliberately baked, they are only preserved in burnt contexts, either destructions or accidental fires. This makes the observation of stages of development difficult. The absence of administrative documents in the Archanes script may simply be due to the absence of the appropriate depositional contexts.

There was a long tradition of the use of stamp seals in the Aegean, so we might expect seals to have been stamped on clay. Stamps are attested in the Aegean as early as the seventh millennium (Early Neolithic), becoming commoner by the sixth millennium (Papathanassopoulos 1996: 331–4; Krzyszkowska 2005: 33–5). By the third millennium BC (Early Bronze Age) direct evidence of their use is widespread in the form of impressed clay pieces (direct-object sealings), preserved by burning, that sealed various types of containers, both on the Greek mainland, at a number of sites in the Peloponnese, and, in smaller numbers, on Crete at the same time (Krzyszkowska 2005: 45–56, 76–8). Essentially the same system is attested in over 6000 stamped clay direct-object sealings preserved in the Proto-Palatial palace of Phaistos as well as several hundred at contemporary Monastiraki (Krzyszkowska 2005: 104–9) (Fig. 1.2). Although cylinder seals are known as artefacts from Pre-Palatial Aegean contexts, the style of sealing actually used on clay in the Aegean was always stamping, not rolling. This practice of stamp sealing further reminds us that administration is a practice that may or may not involve writing, and that accounts of change should include the broadest administrative context, not focus narrowly on its written manifestations (e.g. Palaima 1990).

The second modification to Evans' evolutionary scheme was the discovery of Linear A documents in Proto-Palatial contexts in the palace of Phaistos that were contemporary with the clay sealings mentioned above, long before the 'demise' of Cretan Hieroglyphic (e.g. Schoep 1995: 38–43). In that period, uses of Cretan Hieroglyphic and Linear A on clay documents seem to have had a complementary distribution: in the north and east of the island (particularly the sites of Knossos, Malia, and Petras), Cretan Hieroglyphic was used, while Linear A is attested in the south (currently only Phaistos; e.g. Schoep 1999a). Why both scripts should be in contemporaneous use is unclear. They may notate different languages, although similarities between the two scripts are such that some (e.g. Karnava 2005; Younger 1999: 387–8, 393) regard them as stylistic variants of a single script ('glyptic-derived' versus 'linear'), each embedded in different local administrative systems.

Finally, crucial for this analysis is Evans' chronological positioning of Linear B at Knossos. The recognition that the phase of the palace at Knossos in which the Linear B documents were preserved post-dated destruction levels at other Neo-Palatial sites, rather than being contemporary with them, emerged from the controversy that erupted in the early 1960s about the dating of the tablets (first summarized in Palmer and Boardman 1963). This controversy was sparked in part by a desire to 'tidy up' the cultural sequence in the area. The discovery of Linear B documents at Pylos in Messenia (Fig. 1.1), dating to the end of the Late Helladic IIIB period, c. 1200 BC, together with other discoveries at mainland sites of similar date, left the deposits at Knossos, belonging to a period a couple of centuries earlier, chronologically isolated. The fact that the Pylos docu-

ments were so similar to those at Knossos, combined with Leonard Palmer's scepticism about the quality of Evans' recording of the excavation, led him to propose that the Pylos and Knossos tablets were contemporary, a view vigorously denied by John Boardman. Re-study of the Knossos material by Mervyn Popham (1970) and others, plus newly excavated contexts, demonstrated that the Knossos documents were indeed much earlier than those at Pylos, but belonged to a phase that clearly post-dated the end of the Neo-Palatial period. The situation is quite involved, as one would expect with a large building complex in use over several centuries, but it now seems that the oldest surviving Linear B documents at Knossos were preserved in an area called the 'Room of the Chariot Tablets' (Driessen 2000). These belong early in the LMIIIA1 period (see Table 1.1), perhaps around 1400 BC, while the majority of the rest of the archive dates a generation or two later, in early LMIIIA2, although some small groups may date later still (the situation is summarized in Driessen 1990; 1997a).

Linear A in Use: Neopalatial Minoan Administration

As noted above, the Linear A script was in use on Crete from at least the Middle Minoan IIB phase (c. 1800 BC). In the Neo-Palatial period (c. 1700–1450 BC), Linear A became predominant, and it is the sole script attested by its latest phase. From this period Linear A is also attested at a number of sites off the island, mostly in the Cycladic islands to the north (Karnava 2008).

We know of almost 1500 inscriptions in Linear A (Schoep 2002: 13–22, from which the statistics in this paragraph are drawn). Of these, 113 (7.5%) can be categorized as 'non-administrative', appearing on a range of objects, such as ceramic vessels (both incised and painted), stone vessels (Fig. 1.3), and gold and silver items of personal adornment (Table 1.2). A few of these may have had some relation to economic administration, such as the 42 that were incised before firing on large-scale ceramic storage vessels, including the example from Kato Zakros (Fig. 1.1) with signs numbering at least 24, plus the logogram for 'wine', followed by the figure 32 (ZA Zb 3: Godart and Olivier 1982: 112–13). Others represent longer, continuous text, without logograms or numerals. Among these are the 42 examples incised on stone vessels, many of them deposited in sanctuaries (alongside much larger numbers of uninscribed examples), such as that on the peak of Mt Iouchtas c. 15 km south of Knossos (Fig. 1.2). One vessel from this context bears an inscription of over 37 signs (IO Za 2: Godart and Olivier 1985: 18–19), while another example, from the far-eastern site of Palaikastro (Fig. 1.2), has 45 signs (PK Za 11: Godart and Olivier 1982: 32–5). Rarer are precious-metal objects deposited in tombs, such as a gold ring from the Mavrospelio cemetery at Knossos with 19 signs arranged in a spiral (KN Zf 13: Godart and Olivier 1982: 152–3), or a silver pin from a tomb at Platanos (Fig. 1.2) in south-central Crete, with at least 24 signs (PL Zf 1: Godart and Olivier

1982: 161). Although we may classify these as non-administrative documents, on the basis of material and find contexts, they were certainly bound up with practices of the elite, who may well have seen no distinction between recording on clay and inscribing on metal or stone.

The vast majority of preserved Linear A inscriptions can be considered 'administrative', and were inscribed on clay media specifically shaped to reflect their function within that system: tablets, and seal-impressed clay pieces in a range of forms (Table 1.2). About 1370 inscribed administrative documents are known, of which the majority were preserved in the burnt destructions of various major sites, notably Ayia Triada in south-central Crete, Khania in the north-west, and Kato Zakros on the east coast (Fig. 1.2). They therefore belong at the end of the Late Minoan IB phase, c. 1450 BC (Schoep 1995; Hallager 1996: 39–77). Of 330 known tablets, 147, the largest single group, come from Ayia Triada. This is the only class of document designed exclusively as a support for writing. Linear A tablets occur in only one form, with the proportions of a playing-card, and bear, on average, 16.1 signs per tablet (Schoep 2002: 38).

All other classes of document involve seal-impressions, but do not *require* writing. With one exception (Younger 1999: 384–5), Linear A writing does not appear on seals, unlike the situation with Cretan Hieroglyphic; rather, seals contain pictorial scenes, sometimes of considerable elaboration (e.g. Hallager 1985). By the Neo-Palatial period, all writing is incised or painted; it was not formed by

Figure 1.3 Linear A inscribed stone offering table from cave sanctuary at Psychro (PS Za 2). Ashmolean Museum, Oxford. Max. W. approx. 40 cm.

<table>
<tr><th>Type</th><th>Classification*</th><th>Medium</th><th>Style of mark</th><th>Function</th></tr>
<tr><td>tablet</td><td>[site + no.]</td><td rowspan="5">clay</td><td>incision</td><td rowspan="6">administrative</td></tr>
<tr><td>roundel</td><td>Wc</td><td rowspan="2">incised/stamp</td></tr>
<tr><td>one-hole hanging nodule</td><td>Wa</td></tr>
<tr><td>two-hole hanging nodule</td><td>Wd</td><td rowspan="2">stamp/incised (rare: c. 5%)</td></tr>
<tr><td>nodulus</td><td>We/Wf</td></tr>
<tr><td>flat-based nodule</td><td>Wb</td><td>clay + parchment</td><td>stamp/incised (very rare: < 1%)</td></tr>
<tr><td rowspan="5">vessel</td><td>Zb</td><td rowspan="3">clay</td><td>incised pre-firing</td><td>administrative?</td></tr>
<tr><td rowspan="2">Zc</td><td>painted pre-firing</td><td rowspan="4">non-administrative</td></tr>
<tr><td>painted post-firing</td></tr>
<tr><td>Za</td><td rowspan="2">stone</td><td rowspan="3">incised</td></tr>
<tr><td>architectural block</td><td>Ze</td></tr>
<tr><td>personal adornment</td><td>Zf</td><td>metal</td></tr>
<tr><td>wall</td><td>Zd</td><td>plaster</td><td>administrative?</td></tr>
<tr><td>miscellaneous</td><td>Zg</td><td>various</td><td>incised or painted</td><td>non-administrative</td></tr>
</table>

*For classification of sealed document types (W-), see Hallager 1996: 21–9; Schoep 2002: 16–21; for other types (Z-), see Godart and Olivier 1982; Schoep 2002: 13–16.

Table 1.2 Types of inscription in Linear A, indicating media, function, and style of marking.

the process of stamping. The commonest type of seal-impressed document is a simple clay nodule, formed around the knotted end of a cord, and referred to as a 'single-hole hanging nodule'. Of these 983 are known (936 from Ayia Triada alone), 878 of them bearing inscriptions, almost exclusively comprising a single Linear A sign. At Ayia Triada, these were found in large numbers, concentrated in the northwest quarter of the main building, almost certainly fallen from an upper storey (Hallager 1996: 43; Schoep 2002: 194). The most plausible interpretation offered to date is that these—perhaps in pairs—sealed rolled papyrus documents, which have not themselves survived (Hallager 1996: 159–99, esp. 198–9; Krzyszkowska 2005: 158–60). A second large group of 761 sealings (only seven bearing inscriptions, however) reveal, under microscopic examination, the imprint of animal skin and therefore seem to have sealed tightly folded and bound parchment documents or 'packets', again not preserved (Pini 1983:

560–2; Hallager 1996: 135–58; Krzyszkowska 2005: 155–8). This probable indirect attestation of other writing media reminds us that traditions parallel to that attested on clay must have existed (supported by the existence of longish ink inscriptions on the inside of two cups found at Knossos—KN Zc 6 and 7: Godart and Olivier 1982: 118–25), and that we need to be as much aware of what we are *not* observing as of what we are observing.

Three other types of documents bear seal-impressions. 'Roundels' are known in 180 examples, of which 139 bear inscriptions, usually single signs, it seems used logographically to denote a commodity. As their name suggests, these were small disks of clay, inscribed usually on one face, with varying numbers of identical seal-impressions round the edge. These were probably receipts, held by the issuing party, the number of seal-impressions corresponding to the quantity of the commodity issued (Hallager 1996: 79–120; Krzyszkowska 2005: 163). The two remaining types, *noduli* and two-hole hanging nodules, are rarely inscribed (less than 5% of examples: Table 1.2). *Noduli* are simply small pieces of clay with one or more seal-impressions, depending on their shape. Their exact function is unclear, but may have included that of 'dockets' or 'tokens' (perhaps to allow passage, or to receive accommodation), although some may have been issued to receivers as receipts *from* an authority (Hallager 1996: 121–33; Krzyszkowska 2005: 161–3). Two-hole hanging nodules were formed around a knotted cord that protruded on both sides (hence 'two-hole'); they seem to have tagged or labelled objects, possibly valuables (Hallager 1996: 159–61; Krzyszkowska 2005: 160–1).

Understanding the administrative process to which these documents attest is not easy, given both that the script is undeciphered and that longer, presumably summarizing and potentially 'archival' documents on papyrus and/or parchment are absent. Schoep divides the process into: 'primary' documents, for information-gathering, among which she includes the *noduli*, some tablets, some roundels, and some flat-based nodules (i.e. parchment documents); 'intermediate' documents, for information-processing, chiefly tablets, but also some roundels and some flat-based nodules; and 'final' documents, the papyri implied by the large numbers of single-hole hanging nodules, although she also allows a role here for roundels and flat-based nodules, and perhaps for the two-hole hanging nodules (summarized in Schoep 2002: 193–9, esp. fig. 4.4). Pietro Militello (1991), basing his study on the largest surviving 'system' at Ayia Triada, develops a similar reconstruction, moving from 'data-collection' through 'data-elaboration' to 'data-archiving'. However, he distinguishes processes taking place within the central complex and those that happened outside, reflecting the centre's interaction with a territory. Unlike the situation with Linear B administration, no hierarchy is obvious within the documentation on tablets, although it is clear, for example, from tablets that have multiple topics and formats that they drew data from other sources (Schoep 2002: 194). The

overall picture strongly suggests that we are missing a summary stage that presumably existed originally on the perishable documents.

A total of forty-five sites on Crete, the islands (Thera/Santorini, Melos, Keos, Samothrace), and the Greek (Ayios Stefanos) and Anatolian (Miletos) mainlands have produced Linear A inscribed material (Fig. 1.1), although only eleven of those sites have both administrative *and* non-administrative types. More significantly, sites at either end of Crete—Khania in the west, and Kato Zakros in the east—plus Ayia Triada in south-central Crete, have the full repertoire of administrative document types in the LMIB phase (Schoep 1995). This suggests that the administrative 'reach' of any one of these various centres did not extend over the majority of the island, although the existence of flat-based nodules in non-local clay at some sites implies administrative or diplomatic interaction. That the scale of administration was regional, rather than island-wide, is also supported by the overall figures attested in the better-preserved collections of documents (Schoep 1999b; 2002: 199). A surprising gap in the record of Linear A documents preserved in the LMIB phase is Knossos itself. Although the site produced examples of administrative and non-administrative documents, those that can be given relatively clear contexts—especially the clay documents—belong to earlier phases, mostly LMIA (Schoep 1995: 31–4). The significance of this absence is difficult to gauge, especially given that unbaked clay documents are preserved only where they have been burnt, but if it were real, it could be relevant to the disappearance of Linear A more generally.

Over 1000 inscriptions in Linear A belong to the end of the LMIB phase. It is therefore striking that only three definitively Linear A inscriptions are known from later contexts. Two of these may actually have been manufactured and inscribed in or before LMIB: a ceramic storage jar with six signs incised prefiring on its shoulder that was found in the Unexplored Mansion near the palace at Knossos (KN Zb 40: Godart and Olivier 1982: 83; Popham, Pope, and Raison 1976) is LMII by context, while a handle found in a closed LMII–IIIA deposit at the eastern site of Palaikastro bears a single, incised sign (PK Zb 24: Schoep 1998). The third, and most convincing, example is a clay female figure, LMIIIA1 by style, bearing an inscription painted before firing on her skirt, and recovered from the Knossian harbour site of Poros (PO Zg 1: Dimopoulou *et al.* 1993). None of these three examples can be categorized as administrative documents. A fourth possible example also falls into this category: the two signs incised on an architectural block forming part of the entrance to a tholos tomb (a type of Late Bronze Age tomb, with a circular chamber constructed in corbelled stone blocks, approached by a narrow corridor, or *dromos*: e.g. Cavanagh and Mee 1998: 44–6) at the location named Kephala, north of the palace at Knossos (Preston 2005). Godart and Olivier include this in their catalogue of Linear A inscriptions (KN Ze 16: Godart and Olivier 1982: 138; cf. Preston 2005: 62), but the signs involved are attested in both scripts and the more likely construction date for the

tomb is LMII (Preston 2005: 83–6). If we accept the inscription as Linear A, then it is another example of the continued use of the script in a non-administrative context; if we consider it Linear B, then it would be a highly unusual use.

We should also note, in passing, the puzzling example of a stone plaque dated to the Hellenistic (third century BC) or Roman periods and commonly thought to have been recovered from the Psychro cave sanctuary in east-central Crete, or possibly from the village of Ini. This object bears an alphabetic inscription in the so-called Eteocretan language found mainly in inscriptions of the seventh to third centuries BC, but also has three signs that bear a striking resemblance to the Linear scripts (Duhoux 1982: 95–111, 328, pl. 27). Since a Bronze Age Linear A inscription is known from this site (PS Za 2: Godart and Olivier 1982: 52–5), it is possible that the inscription, if genuine (see Duhoux 1982: 110–11), copied signs from an ancient object encountered in the cave (cf. Alcock 2002: 116–7; Watrous 1996: 55).

If this disappearance of Linear A represented the (almost complete) disappearance of writing on Crete, then the situation would be relatively clear, but the Linear B script is attested there and, in later contexts, on mainland Greece, as we observed above. Moreover, the Linear B script is evidently an adaptation of Linear A, with modifications to the syllabic and logographic repertoires, and to the system of recording weights and measures. The earliest known Linear B documents are from the Room of the Chariot Tablets area of the palatial complex at Knossos (Driessen 1990; 2000), and, although their precise dating remains controversial, they appear to belong two to three generations after the end of LMIB, at the beginning of the LMIIIA phase, perhaps around 1400 BC. Linear B documents are not yet attested anywhere else on Crete until the mid-LMIIIB phase, c. 1250 BC. Moreover, with one striking exception, Linear B inscriptions are confined to the administrative world of tablets and clay sealings.

We seem, then, to be dealing with the 'replacement', within a script community, of Linear A by Linear B, rather than with the former's true disappearance. The replacement raises significant issues, since it is absolute, which is striking when, earlier in Crete's history, Linear A and Cretan Hieroglyphic had been in contemporaneous use.

The Relationship between Linear A and Linear B

Evans' observations about Linear B—that it was a Knossian variant of Linear A—are understandable, because the scripts bear a considerable resemblance to one another: '[t]he occurrence in both of similar sign-groups seems at any rate to prove that the language itself of those who used the one or the other Script was essentially the same' (Evans 1909: 38). However, it had become clear, even before the decipherment, that there were differences. With the decipherment came the realization that one important, underlying difference was language.

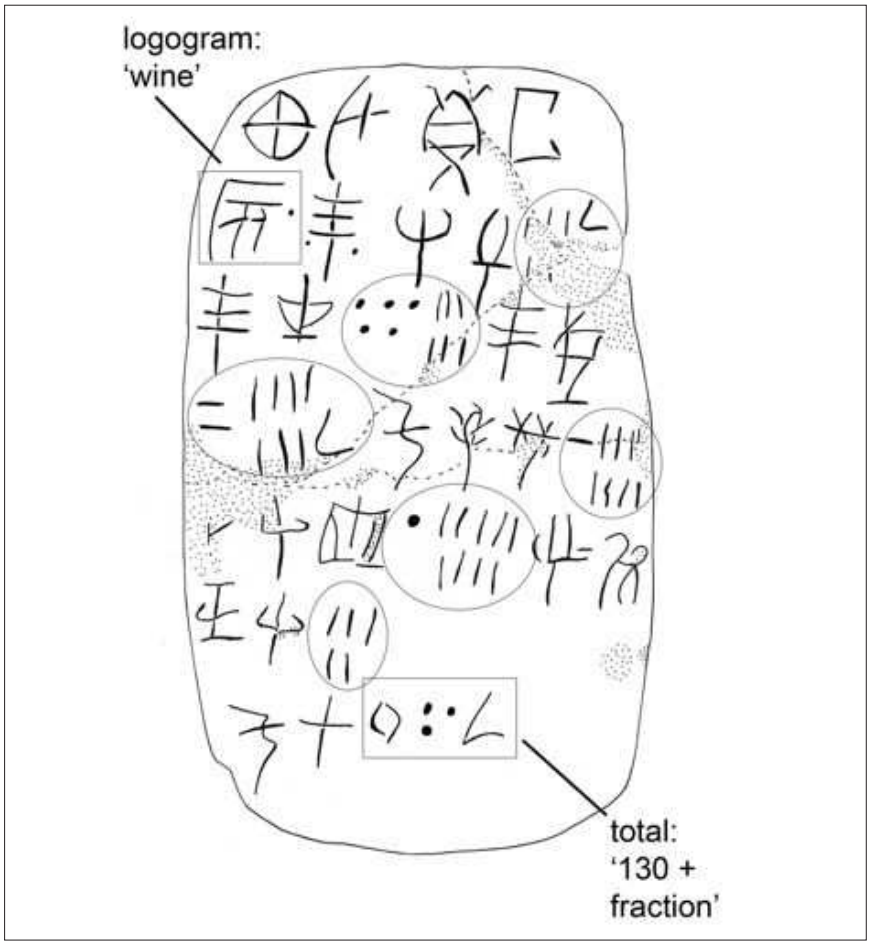

Figure 1.4 Linear A tablet from Ayia Triada (HT 13), annotated to show components of the script. Circles indicate numbers and fractions. H. 10.5 cm.

Relations in script

Both scripts are similar in structure, comprising a relatively small number of syllabograms in comparison to Hittite or Mesopotamian logo-syllabic systems (Figs. 1.4–1.6). In Linear B, the core syllabic repertoire is about 59 signs, while that of Linear A may number as high as 100 (Palaima and Sikkenga 1999: 600). In the case of Linear B, it is clear that the majority of signs are either pure vowels (*a*, *e*, *i*, *o*, *u*) or open consonant (C) + vowel (V) combinations (e.g. *da*, *de*, *di*, *do*, *du*). There are also a few C + glide + V combinations (e.g. *nwa*, *dwe*, *rya*). Nothing in the Linear A syllabary suggests that it was structurally dissimilar, although it appears that the phonetic structure of the language behind Linear

A differed from that of Greek. This is evident when comparing the two sign repertoires: many Linear B syllabic signs are recognizably similar in form to signs in Linear A. Scholars disagree about the exact number of Linear B signs with formal parallels in Linear A, but a figure of 75% overall is plausible; this percentage is even higher (85%) if only the core signs are taken into account (based on Palaima and Sikkenga 1999: 607, table 1). The signs that appear to 'innovate' are not randomly distributed across the grid of Linear B values. Most striking is that seven of thirteen core signs representing C+*o*, and two of the C+*e* series, are 'new' (e.g. Palaima and Sikkenga 1999: 603), while the C+*a*, C+*i*, and C+*u* series all have formal parallels in Linear A. This suggests a different vowel system behind Linear A, most plausibly a three- rather than five-vowel system. In addition to differences in the vowel system, there are some indications that the consonantal system of the language behind Linear A differed from that of Greek. For example, there is a possible *d*/*l* confusion in the Linear A script (Palaima and Sikkenga 1999: 604, n. 16), and Linear A may contain signs for C + glide + V (C^w+V; C^y+V) (Schrijver 2001). Similarly, Linear B appears to have retained some 'inherited' signs whose value cannot yet be determined with certainty, notably Linear B *22 and *56 (see Melena 1987); these possibly reflect sounds in the 'Linear A language', and are particularly common in personal and place-names (Palaima and Sikkenga 1999: 603). The fact that no sign-series in Linear B is entirely new, combined with the probable difference in consonant values, suggest that *some* modification of phonetic values of borrowed signs took place, meaning that great caution should be exercised in using Linear-B-derived values to 'read' Linear A inscriptions (e.g. Younger 2000).

As well as syllabic signs, both Linear A and B used a range of logograms, sometimes referred to as 'ideograms', to denote people, materials, commodities, and objects. Although there is some overlap between the scripts (e.g. Palmer 1995; Schoep 2002: 91–3), Ilse Schoep (2002: 29) estimates that 80% of around 180 Linear A logograms (including multiple variant forms) did not continue into Linear B. The creators and early users of Linear B devised a new logographic repertoire comprising around 140 signs and variants (e.g. Ventris and Chadwick 1973: 50–1), among which are numerous representational or 'iconic' logograms, for example those for 'horse' and 'deer', for vessels, for 'chariots', and for a range of vessel forms, elements of armour, and weapon types. Also Linear B used some syllabic signs in isolation as new logographic signs on the acrophonic principle: e.g. *KO*, abbreviating *koriadna*, 'coriander'; some 'monograms' were created from syllabic signs that spelled words in Greek: e.g. *ME+RI* for *meli*, 'honey' (Ventris and Chadwick 1973: 52–3; Schoep 2002: 29). In the Linear A script logograms are often ligatured—some frequently, such as that conventionally read as 'barley'—with syllabic signs, or fractions (Schoep 2002: 29–30). All of these practices are seen in examples of Linear A logograms that continue into Linear B, such as the cloth logogram with various infixed syllabograms (e.g. Melena 1975: 107–17) or

the 'wool' logogram, which is probably a monogram created in Linear A from the signs which have the syllabic values *ma* and *r/lu* in Linear B, or the logogram for 'figs', which has the syllabic value *ni* in Linear B, possibly an abbreviation for *nikúleon*, a word attested in later Greek lexicographers (e.g. Ventris and Chadwick 1973: 563).

While the system of numerals is identical in Linear A and B (both are decimal, as was Cretan Hieroglyphic, and both use the same symbols for units, tens and hundreds: Schoep 2002: 30–1), the way in which the two scripts handled weights and measures differs radically. Linear A has an extensive range of around 17 'simple' and 30 'composite' signs denoting fractional quantities, conventionally referred to as 'klasmatograms' (Schoep 2002: 31). Thus, a weighed quantity of a commodity might be expressed as '50¾' 'talents'. In Linear B, by contrast, weights and dry and liquid volume measures were notated in three series, using a hierarchy of quantities: major unit, plus four smaller units for weights, and three each for dry and liquid volumes (e.g. Ventris and Chadwick 1973: 54–60, 393–4). Thus, the same '50¾' quantity might be expressed in Linear B, on the lines of pre-metric British weights, as '50 cwt 84 lbs'. Some of the Linear A 'klasmatogram' signs were re-used in this system, but not, as far as we can tell, systematically with the same fractional ratios. This change in practice suggests a revision to accommodate a new, or external, system, seemingly sexagesimal (e.g. Petruso 1978; De Fidio 2002; generally Michailidou 2001).

Relations in administration

At the higher level of administrative practice, differences between the two administrative communities are still clearer (see Palaima 2003 for a concise summary of Linear B practice). The range of documents produced in Linear B is much more restricted (Table 1.3). The only class of texts other than clay-based administrative documents was painted, before firing, on to about 160 examples of a type of transport/storage vessel called the stirrup-jar. The vast majority of

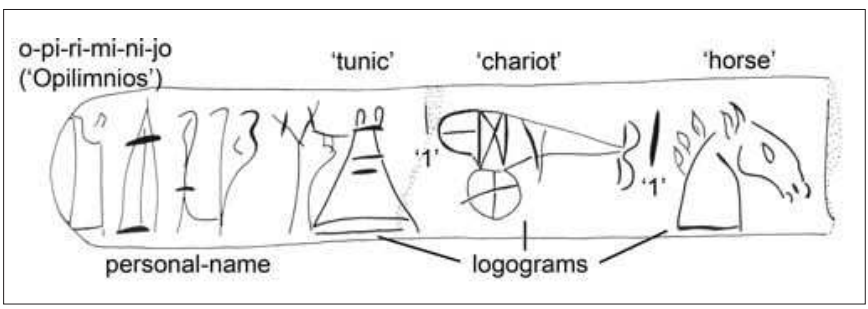

Figure 1.5 Linear B single-entry tablet from the Room of the Chariot Tablets deposit at Knossos (KN Sc 230), indicating combination of phonetic signs, numerals and logograms. W. 13 cm.

<table>
<tr><th>Type</th><th>Classification*</th><th>Medium</th><th>Style of mark</th><th>Function</th></tr>
<tr><td>tablet (page)</td><td>An–z</td><td rowspan="6">clay</td><td rowspan="4">incision</td><td rowspan="7">administrative</td></tr>
<tr><td>tablet (elongated)</td><td>Aa–m</td></tr>
<tr><td>label</td><td>Wa–b</td></tr>
<tr><td>string nodule</td><td>Wm; Wo</td></tr>
<tr><td>regular/irregular string nodule</td><td>Wr–y</td><td rowspan="2">stamp/incision (rare: c. 14%)</td></tr>
<tr><td>nodulus</td><td>Wn</td></tr>
<tr><td>flat-based nodule</td><td>Wb [only RCT at Knossos]</td><td>clay + parchment</td><td></td></tr>
<tr><td>vessel</td><td>Z</td><td>clay</td><td>painted pre-firing</td><td>administrative?</td></tr>
<tr><td>weight?</td><td>Zh</td><td>stone</td><td>incision</td><td>non-administrative?</td></tr>
</table>

* For classification of tablets, see Ventris and Chadwick 1973: 50–1; for sealed document types (W), see Hallager 2005: 249–58.

Table 1.3 Types of inscription in Linear B, indicating media, function and style of marking.

the vessels were not inscribed, and they were probably manufactured on Crete, but were found widely distributed at sites on Crete and the Greek mainland; most date to the later fourteenth and thirteenth centuries (e.g. Van Alfen 1997; Haskell *et al.* in press). There are a handful of inscriptions on stone (including, perhaps, that on the Kephala tholos tomb, mentioned above), or incised on ceramics other than stirrup-jars, but nothing like the quantity attested in Linear A, despite the fact that Linear B written administration is attested not only on Crete but also at a number of mainland sites (Pylos, Mycenae, Tiryns, Midea, and Thebes: Fig. 1.1).

As with Linear A, Linear B administrative documents include tablets and a range of other clay documents, many of them stamped by seals. Among the approximately 5,500 documents attested in Linear B, the vast majority are tablets that exist in two forms, unlike those in Linear A: elongated (or 'palm-leaf'), and page-shaped. The former, a type not attested in Linear A, were used predominantly to record single units of information, single 'rows' of data as in a computer spreadsheet (Fig. 1.5). Page-shaped tablets, on the other hand, had multiple rows of data, sometimes clearly summarizing information recorded elsewhere on individual documents, either elongated tablets or string nodules (Fig. 1.6). There is thus a clear hierarchy and progression of data-recording within the clay media, and scholars sometimes refer to Linear B as more obsessed with paperwork (or 'claywork'); such a hierarchy is far less obvious in Linear A (e.g. Schoep 2002: 193; see also Militello 1991; Driessen 1997b). What is far from clear in Linear B administration is whether there was a top level of summary

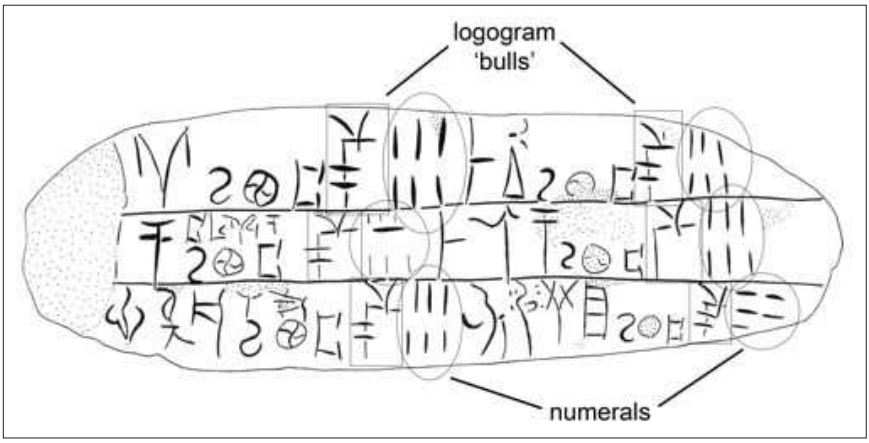

Figure 1.6 Linear B multiple-entry tablet from the Room of the Chariot Tablets deposit at Knossos (KN Ce 59), indicating 'working' male oxen at six places in central and western Crete. W. 14 cm.

recording on perishable material, although many scholars assume that this was so (e.g. Palaima 2003: 171–2; Driessen 1997b; contrast Bennet 2001). There are only seven examples of flat-based nodules, all from the Room of the Chariot Tablets deposit at Knossos, the earliest attested Linear B (e.g. Driessen 1990; 1997a; 2000), although they differ in detail from Linear A examples, leading to the suggestion that they might not have sealed parchment documents like their Linear A counterparts (Hallager 2005: 252). No single-hole hanging-nodules are known, implying that papyrus documents, if they existed, were not sealed in the same way as they had been in the Neo-Palatial period.

Equally, the sealing system of Linear A administration is simplified in Linear B (see Hallager 2005 for an excellent summary): roundels disappear altogether, while *noduli*, only in one type, become very rare. Single-hole hanging-nodules also disappear, as noted above, while the two-hole variety is split into two types: regular and irregular string nodules. Of these, only the former bear inscriptions, in about 50% of preserved examples. Regular string nodules were probably attached to commodities both within and outside the palatial centres, while the irregular examples (all found broken, probably in antiquity) may have been attached to commodities that had arrived in the palatial buildings themselves, and were broken when no longer required (Hallager 2005: 257–8).

In terms of 'administrative reach', the Linear B documents seem to operate at a higher level, often recording large quantities, either individually or collectively across sets of documents. The loss of the roundel might reflect a distancing of the administration from those to whom it issued goods. The clearest evidence, however, is a product of the decipherment: the fact that we can read and recognize some place-names. These suggest that the 'reach' of Knossos

extended as far as Khania in western Crete, over much of west-central and central Crete (including the region of Ayia Triada and Phaistos), to the east-central part of the island, if probably not to the far east (e.g. Bennet 1985; 1987; 2009). It is possible that the extent of this reach might not have been as great at the time of the earliest administration attested in the Room of the Chariot Tablets deposit, nor might control have been continuous over a single, undifferentiated territory (e.g. Driessen 2001). However, the Linear B documents confirm, for the first time in Crete's prehistory, that Knossos exercised some form of control over the greater part of the island.

Knossos and Crete in the Fifteenth Century BC: The Disappearance of Linear A

It is clear from the above that a new script (Linear B) and administrative system was created at some point before its attestation around 1400 BC in the Room of the Chariot Tablets within the palace at Knossos. This observation raises four questions: how, when, where, and why was Linear B created? The first we have largely answered above in outlining the relationship between the two scripts and their related administrative practices. We can give a *terminus ante quem* to the second question: no later than Linear B's earliest attestation, c. 1400 BC. A more considered date can only be proposed if the third question is answered first.

There are broadly two possibilities for the locus of creation of Linear B: mainland Greece, among a community of Greek speakers, or on Crete, probably at Knossos itself, in an environment, probably bilingual, where Greek was either prestigious or strategically advantageous. In the first instance, the adaptation can be seen as part of the appropriation of exotic materials and knowledge by the early Mycenaean elite, most prominent in the period of the Shaft Graves at Mycenae (see Wright 1995; Palaima 1995; Bennet 2004). It is easy to see how the mid-twentieth-century view of identity, with its emphasis on language, would favour the creation of a Greek-based administration in a Greek-speaking milieu, and its subsequent introduction to Crete by conquering 'Mycenaean' mainlanders. However, with a single possible exception discussed below, there is no evidence for writing on the Greek mainland prior to the LHIIIA2 (mid to late fourteenth century BC) tablet recently discovered in the Petsas House at Mycenae (Shelton 2006). There are fairly numerous examples of Cretan seals deposited in early Late Bronze Age mainland tombs, often in pairs (Rehak and Younger 2000), but no writing. Indeed, it has been argued that the need for complex administration only arose with the earliest palatial structures on the mainland (e.g. Palaima 1988: 336–42), perhaps around the same time as the earliest attested examples of Linear B at Knossos.

Why should the converse be more likely, however: namely the creation of a Linear B script and administration at Knossos on Crete? First, the Linear A and B scripts and administrative systems show considerable similarities, and indeed they seem to diverge with time, the Room of the Chariot Tablets system representing a closer approximation to Linear A practice than later patterns (e.g. Driessen 2000: 224–8). It is not difficult to imagine a situation in which bilingual administrators, in the cosmopolitan major Aegean centre that Knossos was in the second millennium BC, could have adapted Linear A to record the Greek language. Modifications to the administrative system would have been made to reflect the recently enlarged area administered from Knossos—essentially moving administrative activity up to a higher level—and also the centralization of administrative control at Knossos itself.

In support of this scenario, we can first note that we have no evidence for the use of Linear A at Knossos in the immediately preceding LMIB period, although this may simply be a taphonomic issue: no burnt contexts at the site that would have preserved unbaked documents. If so, it is surprising that Linear A documents of all types are so widely attested elsewhere, even if in very small numbers at most individual sites. It is possible, therefore, that a new script may have been adapted from Linear A already in LMIB. Second, we observe quite profound changes in burial practice and material culture, centred on Knossos itself, beginning in the LMII period. Laura Preston has summarized the burial practices, which include new tomb types, especially built examples, like the Kephala tholos tomb (Preston 2005), and new locations for burial, in prominent ridge-top locations north of the palace, on routes to its harbour-town at Poros (Preston 1999). Although the new tomb types have been seen as mainland-derived, and therefore as further evidence for mainland incursions, their range of resonances is more complex, indeed drawing on mainland themes and forms, but also combining them with local Cretan symbols, or even materials, in the case of the re-used ashlar blocks in the Kephala tholos (Preston 1999; 2005). Similarly, new vessel types, particularly the stemmed drinking cup, imply a shift in styles of display on commensal occasions. It is even possible to read the application of an inscription to the entrance of the Kephala tholos as drawing on a Cretan tradition of inscribing on stone and using 'mason's marks'. Moreover, if we were to read the inscription as Linear B, and therefore Greek, it reads *a-pi*, a possible imperative form, 'Go away!' (see Nagy 1963: 205).

We can suggest that the cultural changes at Knossos that began with the LMII period and included the adaptation of administration to form a new system were part of an active construction of a new identity for the Knossos elite on a pan-Cretan, possibly an Aegean stage (see also Driessen 2002; Driessen and Langohr 2007). Similar situations might be the shifts in administrative practice, and language, introduced in Mesopotamia in the Sargonic period in the mid third millennium BC, and then again with the establishment of the Ur III

dynasty a century or so later (e.g. Van de Mieroop 2007: 63–84; Michalowski 1991; Postgate 1994: 40–3).

The ease with which an administrative shift such as this could have been achieved becomes clearer when we consider the likely size of the script-using communities involved. Analysis of scribal hands in the Ayia Triada Linear A documents suggests a possible minimum number of tablet-writers of eighteen (e.g. Godart and Olivier 1985: 83–6; Schoep 1996: 335–55). More 'hands' are identified on the single-hole hanging nodules, but it is difficult to estimate how many individuals this represents, since these are almost all single-sign inscriptions. In the Knossos Room of the Chariot Tablets, Driessen identifies nine major and four minor Linear B 'scribes' responsible for about 650 surviving documents (Driessen 2000: 71–99), against a possible total of around sixty for the much larger corpus attested elsewhere in the palace of Knossos and in later contexts (Driessen 2000: 150–1; see also Olivier 1967). If we accept the idea that Aegean writers were not merely secretaries, but were themselves active in administration (Bennet 2001; see also Palaima 2003: 175–7), then the possibility of a deliberate and active adaptation of the script is still more likely. Moreover, there is no reason for this to have taken a long time (e.g. Palaima 1988: 341); it could certainly have been accomplished within a generation. (Achieving administrative control over much of Crete may have taken longer, of course.) The hypothesis of a relatively quick creation also militates against the idea of a longer 'gestation' on the mainland, prior to the script's introduction, fully formed, to Crete.

There is one small, but potentially significant, piece of evidence against the suggestion that Linear B was adapted on Crete: the discovery, on 1 April 1994, of a small pebble bearing a Linear B inscription at the site of Kafkania, near Olympia in the northwestern Peloponnese (Arapojanni *et al.* 2002). The context for this object is given as late Middle Helladic, perhaps c. 1600 BC (earlier on the high chronology: Table 1.1), but certainly at least two centuries earlier than the Room of the Chariot Tablets at Knossos. The object is peculiar: unique in form and style of inscription, and in its early date, as well as being on stone (but see Adrimi-Sismani and Godart 2005: 59–65), not to mention its place of discovery, well away from the 'heartland' of early Mycenaean developments in the Argolid and Messenia, respectively in the northeastern and southwestern Peloponnese. Thomas Palaima has outlined these and further difficulties with the inscription, drawing attention, among other things, to its precocious sign-forms, which resemble closely Pylos 'hands' some 400 years later in date (Palaima 2006). Although this object cannot be ignored, so long as it remains unique, it cannot be used by itself to overturn the wealth of evidence for the later adaptation of Linear B, whether this occurred on the mainland or, as I argue, on Crete.

If we accept the probability that Linear B was created at Knossos on Crete, then the date is likely to be between c. 1450 and 1400 BC. As to why it was created, I would argue that it was part of a major cultural realignment among the

Knossian ruling elite (including those who used writing), no doubt bilingual, who chose to differentiate themselves through a range of cultural materials and practices drawing on both local and mainland traditions. This realignment included the adaptation of a script that was already at least 300 years old (and so perhaps distanced from spoken forms of its language) to a new written language. It also involved the redesign of indigenous administrative practice for a new context, in which Knossos was expanding its control on the island, control it was to exercise for a further seventy-five or 100 or so years. Seen in this perspective, Linear A did not simply disappear; rather, it was killed. Mid-twentieth-century attitudes to ancient identity and its essentialist link to language led to mainland Greek speakers ('Mycenaeans') being 'framed' for this murder. More probably the guilty party was within the family, so to speak. We can better appreciate the mid-fifteenth-century BC changes on Crete as telling us about how elite notions of identity were constructed and played out in public, than by trying to read modern concepts of identity, shaped by our belonging within modern nation-states, into the material record of a period that is as 'foreign' as any other in the distant past.

Note

1. I should like to thank my fellow editors for proposing and helping to organize the original meeting in Oxford, and for their extraordinary patience in waiting for this chapter; both also offered valuable comments, some at long distance, using the latest technology. I am also grateful to Paul Halstead and Debi Harlan for reading and commenting on the final version, as I am to Artemis Karnava and Peter Schrijver for sharing copies of material in advance of publication. None of them should be held responsible for the use to which I have put—or not put—their advice. Finally, I would also like to thank the staff at Keble College, Oxford, for creating such a pleasant physical environment for our meeting and my former colleagues there for offering a stimulating intellectual environment during the tenure of my fellowship.

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